

Mrs. Washington, the school guidance counselor, at Ready High School is concerned about the growing number of students with jobs (part-time and/or full time). A school wide survey was conducted asking students about their job status. 1074 of Ready High School's 2000 students have some type of job.

Students in all grades were surveyed (9^{th} , 10^{th} , 11^{th} , and 12^{th}). The possible responses were Yes (part-time or full-time job) or No (no job).

- 23.5% of students surveyed were in the 9^{th} grade
- 27.6% of students surveyed were in the 12^{th} grade
- Of students without a job, 30.5% are 11^{th} grade students.
- Of students surveyed, 20.8% were in 10^{th} grade.
- 20.4% of 9^{th} grade students have a job.
- 45.4% of students with a job are in the 12^{th} grade

Create a relative two-way frequency table below for questions 1-4.
Round all answers to the nearest tenth of a percent.

	Part – Time / Full Time Job		
Grade	Yes	No	Total
1. 9^{th}	4.8%	18.6%	23.5%
2. 10^{th}	10.5%	10.3%	20.8%
3. 11^{th}	14.0%	14.1%	28.1%
4. 12^{th}	24.4%	3.2%	27.6%
Total	53.7%	46.3%	100%

Data was collected on the amount of time High School freshmen and seniors spend on social media each day. Students were surveyed and asked about 2 different intervals of daily social media time (0 – 4 hours OR Over 4 hours).

- 47.52% of students in the sample were freshmen.
- Of students that spend 0 – 4 hours each day on social media, 38.96% were seniors.
- 52.15% of students in the sample spent over 4 hours each day on social media.
- 56.28% of freshmen spend over 4 hours each day on social media.

Use this information to answer questions 5-8. Fill in the blank with either joint, marginal, or conditional.

5. 47.52% represents a marginal relative frequency.

6. 38.96% represents a conditional relative frequency.

7. 52.15% represents a marginal relative frequency.

8. 56.28% represents a conditional relative frequency.

9. Do you have enough information to create a two-way relative frequency chart from the information provided? Explain.

Yes, because a partial table can be made from the information given, and then the rest of the table can be filled in.

10. How can you distinguish between a joint relative frequency and a conditional relative frequency on a two-way frequency table? Explain.

Joint relative frequency is the ratio between a categorical value and the total, whereas conditional relative frequency is the ratio of joint relative frequency to marginal relative frequency.